

DO THAI BINH

SOFTWARE ENGINEER | AI & FULL-STACK DEVELOPMENT

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in Full-stack Development, AI/LLM integration, and IoT systems. Experienced in building AI-powered applications using LLM, RAG, and document processing pipelines, as well as developing scalable web applications and backend services. Proficient in Python, Node.js, Angular, C#, and database technologies, with a strong interest in building practical AI-driven software solutions.

Core Expertise: Full-stack Development · LLM & RAG · AI Integration · Backend Engineering · System Integration

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, C#, C/C++, Java
Frontend Development: Angular, HTML5, CSS3, Bootstrap, JavaScript, TypeScript
Backend Development: Node.js, Express.js, ASP.NET Core, Entity Framework, FastAPI, RESTful API, Third-party API Integration
AI & Machine Learning: LLM, RAG, Qwen, LangChain, Prompt Engineering, Embedding Models, Vector Database, Document Processing, OCR, YOLOv8, Logistic Regression, MLP
Databases: MongoDB, SQL Server, MySQL, Supabase
DevOps & Tools: Docker, Linux, Git, n8n, Postman, Playwright, Figma, StarUML
Additional Technologies: MQTT, ESP32, IoT Integration, AVEVA E3D, Unified Engineering, Navisworks, Software Add-in Development

WORK EXPERIENCE

AI & Software Engineer | PEMOSE **08/2026 – Present**

- Researched and developed **AI-powered solutions integrated with engineering software**, including AVEVA E3D, Unified Engineering, and Navisworks.
- Developed software integration systems and custom **Add-ins** to extend engineering software capabilities and automate workflows.
- Built **RAG-based chatbot** solutions for information retrieval and interaction with engineering documents.
- Researched and implemented **automated conversion of P&ID drawings** (CAD, PDF, and image formats) into AVEVA Unified Engineering.
- Worked on AI applications and document processing solutions for oil & gas piping engineering.

Tech Stack: *Python, .NET, AI/LLM, RAG, API Integration, AVEVA E3D, Unified Engineering.*

Full-stack Developer | Antbuddy **05/2026 – 07/2026**

- Maintained and enhanced a **CRM system** using Node.js, Angular, and MongoDB; fixed bugs and implemented new features based on business requirements.
- Developed and integrated **RESTful APIs** with MongoDB.
- Optimized application performance through database query and API improvements.
- Integrated third-party chat platforms and messaging services.
- Developed automated end-to-end tests using **Playwright**.

Tech Stack: *Node.js, Express.js, Angular, TypeScript, MongoDB, Playwright, Git.*

IoT Intern | Koson Technology **01/2026 – 04/2026**

- Researched and developed **Android applications integrated with IoT devices** for the MIQ Resident Management System.
- Conducted functional testing of IoT modules and analyzed system workflows.
- Prepared technical documentation and researched IoT-based resident management solutions.

Tech Stack: *Android, Java, IoT, ESP32, REST API.*

SELECTED PROJECTS

1. AI-Powered Learning Management & Assessment Platform *| Role: Full-stack Developer / AI Engineer*

- Developed and deployed an AI-powered learning platform for elementary school students, focusing on Mathematics and Informatics.
- Built a **RAG chatbot using Qwen 7B** to retrieve and answer questions from uploaded textbooks, learning materials, and sample questions, including relevant images and document references.
- Implemented **conversational exam generation**, enabling teachers to create customized assessments through natural language prompts.
- Developed online examination, automatic grading, and AI-generated feedback features for students.
- Integrated mathematical formula processing and LaTeX rendering for AI-generated questions and answers.

Tech Stack: *Python, Qwen 7B, LLM, RAG, LangChain, Vector Database, Embedding Models, LaTeX, Docker.*

2. Smart Math Whiteboard – AI Handwritten Math Recognition *| Role: Full-stack Developer*

- Developed a desktop application for handwritten mathematical expression recognition, equation solving, and interactive graph visualization.
- Built an **OCR-to-LaTeX pipeline** using Pix2Text and integrated **SymPy** for symbolic mathematical computation.
- Developed a **hybrid AI assistant** combining deterministic mathematical processing with a local LLM.
- Optimized graph rendering and expression processing through caching, reducing refresh time from **1–7 seconds to approximately 0.15–0.5 seconds**.
- Delivered an offline application with approximately **12,000 lines** of C# and Python code.

Tech Stack: *C#, .NET 8, Python, FastAPI, Pix2Text, SymPy, PyTorch, llama.cpp, DeepSeek-R1-Distill-Qwen.*

3. Smart Parcel Locker System for Last-Mile Delivery *| Role: Full-stack Developer / IoT Engineer*

- Designed and developed an end-to-end IoT-based parcel locker system integrating backend APIs, Android applications, and ESP32 devices.
- Implemented **MQTT** communication for real-time interaction between IoT devices and backend services.
- Integrated AI-based face recognition using **InsightFace** for secure parcel pickup.
- Developed a **Decision Tree** model to forecast next-day parcel volume for resource planning.

Tech Stack: *C#, ASP.NET Core, SQL Server, Android (Java), ESP32, C++, Python, MQTT, Docker, InsightFace.* GitHub

4. AI-Powered Academic Consultation Chatbot *| Role: AI Developer*

- Developed a **RAG-based chatbot** using university documents to provide context-aware academic consultation.
- Implemented document retrieval, query optimization, and persistent conversation memory.
- Integrated **Supabase Vector Database** and LangChain to improve contextual retrieval and multi-turn conversations.

Tech Stack: *Python, FastAPI, LangChain, Ollama, Llama 3.2, Supabase, JavaScript.*

EDUCATION

University of Technology and Education – The University of Danang **GPA: 3.7/4.0**

Software Engineering | Major: Information Technology | Expected to graduate with Excellent Classification.

HONORS & PUBLICATIONS

- **First Author – IEEE International Publication:** Design and Implementation of an IoT-Based Smart Parcel Locker System for Last-Mile Delivery Using MQTT and Machine Learning Model (2026). ieeexplore.ieee.org/document/11628354
- **IEEE International Publication:** Design of Stroke Risk Warning System Based on Internet of Things (IoT) Technology and Machine Learning Model (2025). ieeexplore.ieee.org/document/11142086
- **First Prize** – Faculty Scientific Research Competition in Digital Technology (2025).
- **Innovation for Community Health Award** – Viettel IDC Group (2025).
- **Consolation Prize** – University-level Scientific Research Competition, The University of Danang (2025).
- **UTE Challenge Scholarship** (2024–2025); **Excellent Academic Encouragement Scholarships** (2023–2024, 2025–2026).